DEVELOPER ROLE

100 Multiple-Choice Practice Questions

Topics: Java & OOP, Data Structures and Algorithms, SQL/DBMS, Web & HTTP, Spring Boot, Security, Git/Testing, OS and System Design. Select one best answer for each question.

Instructions: 100 questions · 1 mark per question · Suggested time: 75 minutes · No negative marking in this practice set.

1. Which OOP principle bundles data and methods while restricting direct access to internal state?

A. Inheritance
B. Encapsulation
C. Polymorphism
D. Abstraction

2. Which Java keyword prevents a class from being subclassed?

A. static
B. volatile
C. final
D. private

3. Which method is the standard entry point for a Java application?

A. run()
B. start()
C. main()
D. init()

4. What is the default value of an instance variable of type int in Java?

A. 0
B. null
C. -1
D. Undefined garbage value

5. Which collection does not allow duplicate elements?

A. ArrayList
B. LinkedList
C. HashSet
D. Vector

6. Which Java collection maintains insertion order and allows duplicates?

A. HashSet
B. ArrayList
C. TreeSet
D. HashMap

7. What is method overloading?

A. Same method name with different parameter lists
B. Same signature in unrelated classes
C. Replacing a superclass method
D. Hiding a field

8. What is method overriding?

A. Defining multiple constructors
B. Providing a subclass implementation of an inherited method
C. Changing a variable type
D. Calling a static method

9. Which checked exception is commonly raised when a file cannot be found?

A. NullPointerException
B. ArithmeticException
C. FileNotFoundException
D. ArrayIndexOutOfBoundsException

10. Which keyword refers to the current object in Java?

A. super
B. this
C. self
D. current

11. Which Java type stores key-value pairs?

A. ArrayList
B. HashMap
C. HashSet
D. PriorityQueue

12. What does JVM stand for?

A. Java Variable Manager
B. Java Virtual Machine
C. Joint Version Model
D. Java Verified Module

13. Which statement about Java interfaces is correct?

A. A class can implement multiple interfaces
B. An interface can never contain default methods
C. A class can extend multiple classes
D. Interfaces must have constructors

14. What does the static keyword indicate for a field?

A. One copy per object
B. One class-level member shared by instances
C. The field cannot be read
D. The field is always immutable

15. Which class is immutable in standard Java?

A. StringBuilder
B. StringBuffer
C. String
D. ArrayList

16. What is the purpose of a constructor?

A. Destroy an object
B. Initialize a newly created object
C. Handle exceptions
D. Start a thread automatically

17. Which access modifier provides the broadest visibility in Java?

A. private
B. protected
C. default/package-private
D. public

18. Which block is normally used to release resources even when an exception occurs?

A. throw
B. finally
C. catch only
D. assert

19. What does the equals() method commonly test when properly overridden?

A. Reference identity only
B. Logical equality of objects
C. Object size
D. Class loading order

20. Which keyword is used to explicitly raise an exception?

A. throws
B. catch
C. throw
D. final

21. What is the time complexity of accessing an array element by index?

A. O(1)
B. O(log n)
C. O(n)
D. O(n log n)

22. Which data structure follows LIFO order?

A. Queue
B. Stack
C. Heap
D. Graph

23. Which data structure follows FIFO order?

A. Stack
B. Tree
C. Queue
D. Set

24. What is the average time complexity of lookup in a well-distributed hash table?

A. O(n)
B. O(log n)
C. O(1)
D. O(n log n)

25. Which algorithm is typically suitable for finding a shortest path in an unweighted graph?

A. Dijkstra with negative weights
B. Breadth-first search
C. Depth-first search only
D. Bubble sort

26. Which traversal of a binary search tree returns keys in sorted order?

A. Preorder
B. Postorder
C. Inorder
D. Level order

27. What is the worst-case time complexity of binary search on a sorted array?

A. O(1)
B. O(log n)
C. O(n)
D. $O(n^2)$

28. Which sorting algorithm is generally stable in its standard implementation?

A. Merge sort
B. Heap sort
C. Selection sort
D. Quick sort in-place

29. Which data structure is commonly used to implement recursion internally?

A. Queue
B. Stack
C. Hash table
D. Disjoint set

30. What is the worst-case time complexity of quicksort?

A. O(log n)
B. O(n)
C. O(n log n)
D. $O(n^2)$

31. Which technique solves a problem by combining solutions to overlapping subproblems?

A. Dynamic programming
B. Linear probing
C. Binary search
D. Topological sorting

32. A graph with V vertices and E edges is commonly represented using an adjacency list in what space?

A. $O(V^2)$ always
B. O(V + E)
C. $O(E^2)$
D. O(log V)

33. Which data structure is most appropriate for repeatedly retrieving the highest-priority item?

A. Stack
B. Queue
C. Priority queue
D. Linked list

34. What is the time complexity of inserting at the head of a singly linked list when the head pointer is known?

A. O(1)
B. O(log n)
C. O(n)
D. $O(n^2)$

35. Which approach is a common way to detect a cycle in a linked list with O(1) extra space?

A. Binary search
B. Floyd's slow and fast pointers
C. Merge sort
D. Counting sort

36. What is a primary advantage of a balanced binary search tree?

A. Always uses O(1) memory
B. Keeps search, insertion, and deletion around O(log n)
C. Stores only strings
D. Requires sorted input

37. Which SQL command retrieves rows from a table?

A. GET
B. SELECT
C. FETCH TABLE
D. OPEN

38. Which clause filters rows before grouping in a SELECT query?

A. HAVING
B. ORDER BY
C. WHERE
D. GROUP BY

39. Which clause filters groups after GROUP BY?

A. WHERE
B. HAVING
C. LIMIT
D. DISTINCT

40. Which key uniquely identifies each row in a relational table?

A. Foreign key
B. Candidate index only
C. Primary key
D. Duplicate key

41. What does a foreign key primarily enforce?

A. Sorting
B. Referential integrity
C. Encryption
D. Compression

42. Which join returns only rows with matching values in both tables?

A. LEFT JOIN
B. FULL OUTER JOIN
C. CROSS JOIN
D. INNER JOIN

43. What does COUNT(*) count?

A. Only non-null values in one column
B. All rows in the result set
C. Only distinct rows
D. Only primary keys

44. Which SQL statement changes existing rows?

A. ALTER
B. UPDATE
C. CREATE
D. GRANT

45. Which property of ACID means a transaction is all-or-nothing?

A. Consistency
B. Isolation
C. Atomicity
D. Durability

46. What is database normalization mainly intended to reduce?

A. Query syntax
B. Redundancy and update anomalies
C. Indexes
D. All joins

47. Which index structure is commonly used by relational databases for range queries?

A. B-tree/B+ tree
B. Stack
C. Bloom filter only
D. Binary heap

48. What is the purpose of GROUP BY?

A. Sort rows alphabetically
B. Aggregate rows sharing the same grouping values
C. Delete duplicate records
D. Rename a table

49. Which command removes all rows while retaining the table structure and typically supports no WHERE clause?

A. DROP
B. TRUNCATE
C. ALTER
D. RENAME

50. What is a database transaction?

A. A database user account
B. A logical unit of work that is committed or rolled back
C. A type of index
D. A stored file format

51. Which HTTP method is generally intended to retrieve a representation without changing server state?

A. POST
B. PATCH
C. GET
D. DELETE

52. Which HTTP status code commonly indicates successful resource creation?

A. 200
B. 201
C. 301
D. 404

53. Which HTTP status code indicates that authentication is required or has failed?

A. 400
B. 401
C. 403
D. 500

54. What does REST primarily describe?

A. A programming language
B. An architectural style for networked applications
C. A database engine
D. A CSS framework

55. Which format is commonly used for JSON API request and response bodies?

A. application/json
B. text/sql
C. image/java
D. application/xml-only

56. What is the role of DNS?

A. Encrypt HTTP traffic
B. Translate domain names to IP addresses
C. Assign database primary keys
D. Compile Java

57. Which protocol provides reliable, ordered byte-stream delivery?

A. UDP
B. TCP
C. ICMP
D. ARP

58. What does HTTPS add to HTTP?

A. TLS-based transport security
B. Automatic database caching
C. Guaranteed zero latency
D. Compression in every case

59. Which status code most directly represents a server-side internal error?

A. 201
B. 302
C. 404
D. 500

60. In a typical three-tier web application, which layer contains business rules?

A. Presentation layer
B. Business/service layer
C. Database storage layer only
D. DNS layer

61. What is CORS designed to control?

A. Database schema migrations
B. Cross-origin browser requests
C. CPU scheduling
D. Password hashing algorithms

62. Which HTTP method is commonly used for a partial resource update?

A. GET
B. PATCH
C. TRACE
D. OPTIONS

63. What is the purpose of API pagination?

A. Encrypt payloads
B. Return large result sets in manageable chunks
C. Prevent all errors
D. Replace authentication

64. Which Java framework annotation commonly marks a REST controller in Spring?

A. @RestController
B. @EntityManager
C. @AutowiredOnly
D. @ServletTable

65. In Spring, what is dependency injection?

A. Manually creating every dependency inside a class
B. Providing an object with its required dependencies externally
C. Converting Java to SQL
D. Encrypting method names

66. Which annotation maps HTTP GET requests in Spring MVC?

A. @PostMapping
B. @PutMapping
C. @GetMapping
D. @DeleteMapping

67. What is the main purpose of a DTO in an API?

A. Represent the exact database engine
B. Transfer data between application boundaries
C. Replace all services
D. Start application threads

68. Which layer should usually contain HTTP request mapping logic?

A. Controller
B. Repository
C. Database index
D. Build tool

69. What is the role of a repository in a typical Spring application?

A. Render CSS
B. Abstract data access operations
C. Configure DNS
D. Manage browser history

70. What does @Transactional typically help manage?

A. Database transaction boundaries
B. HTML layout
C. JWT signing keys only
D. Network routing tables

71. Which configuration file is commonly used for Spring Boot properties?

A. pom.lock
B. application.properties
C. index.css
D. routes.sql

72. What is the purpose of a database migration tool such as Flyway?

A. Generate UI icons
B. Version and apply database schema changes
C. Load balance HTTP traffic
D. Compress Java bytecode

73. Which password storage approach is recommended?

A. Plain text
B. Reversible Base64 encoding
C. A slow salted password hash such as BCrypt or Argon2
D. A single unsalted MD5 hash

74. What is the purpose of a JWT signature?

A. Hide all token contents by default
B. Help verify token integrity and authenticity
C. Store passwords in plaintext
D. Make tokens impossible to expire

75. Where should production secrets normally be stored?

A. Committed source code
B. Public README
C. Secret manager or protected environment configuration
D. Frontend JavaScript bundle

76. What is SQL injection?

A. A query optimization technique
B. An attack that manipulates SQL through untrusted input
C. A database backup method
D. A form of table normalization

77. Which practice helps prevent SQL injection?

A. String-concatenating user input into SQL
B. Using parameterized queries/prepared statements
C. Disabling indexes
D. Returning database errors to everyone

78. What does the principle of least privilege mean?

A. Give every user admin access
B. Grant only permissions required for the task
C. Disable all authentication
D. Use one shared account

79. What is the main purpose of Git?

A. Run production servers
B. Track changes to files and coordinate version control
C. Store relational data
D. Design web pages

80. Which Git command creates a new local branch and switches to it in modern Git?

A. git reset --hard
B. git switch -c branch-name
C. git stash pop
D. git fetch --delete

81. What does git pull typically do?

A. Only deletes local branches
B. Fetches remote changes and integrates them into the current branch
C. Creates a new repository on every host
D. Rewrites every commit message

82. What is a merge conflict?

A. A successful build
B. Competing changes Git cannot automatically reconcile
C. A deleted remote repository
D. A password mismatch

83. What is the purpose of a pull request?

A. Request review and integration of proposed changes
B. Replace all tests
C. Reset the production database
D. Install an operating system

84. What is continuous integration (CI)?

A. Manually deploying once a year
B. Frequently integrating changes with automated build and test checks
C. Writing code without version control
D. A database replication protocol

85. Which testing level focuses on a small, isolated unit such as a method or class?

A. Unit testing
B. End-to-end testing
C. Acceptance testing only
D. Exploratory production testing

86. What does a mock object help a unit test do?

A. Replace the compiler
B. Simulate a dependency and control its behavior
C. Increase production database size
D. Encrypt test reports

87. What is regression testing?

A. Testing only brand-new features
B. Checking that existing behavior still works after changes
C. Testing network cables
D. Removing old test cases

88. What is the primary purpose of code review?

A. Guarantee no bugs can ever exist
B. Improve correctness, maintainability, and shared understanding
C. Replace all automated tests
D. Avoid documentation permanently

89. Which operating-system component allocates CPU time among runnable processes or threads?

A. Scheduler
B. Linker
C. Parser
D. Load balancer

90. What is a deadlock?

A. A fast context switch
B. A state where processes wait indefinitely for resources held by one another
C. A successful transaction
D. A cache hit

91. Which memory area commonly stores dynamically allocated objects in Java?

A. Heap
B. CPU register only
C. Method call stack only
D. Network buffer only

92. What is a thread?

A. A separate program stored on disk
B. A unit of execution within a process
C. A database table
D. A DNS record

93. What is caching intended to improve?

A. Repeated access latency by keeping reusable data closer to the consumer
B. Source code readability only
C. Password entropy
D. Database normalization

94. What is horizontal scaling?

A. Increasing CPU/RAM on one machine
B. Adding more machines or instances to share workload
C. Removing replicas
D. Reducing all database indexes

95. What is a load balancer used for?

A. Distribute incoming traffic across backend instances
B. Compile TypeScript
C. Store passwords
D. Replace application monitoring

96. Why should an API be idempotent where appropriate?

A. Every request must create a new record
B. Repeating the same request should have the same intended effect
C. It must never return errors
D. It disables authorization

97. What is observability commonly built from?

A. Logs, metrics, and traces
B. Only screenshots
C. Only source comments
D. CSS, HTML, and fonts

98. What is the main benefit of using environment-specific configuration?

A. Hard-code production secrets
B. Keep deployment settings separate from application code
C. Remove the need for testing
D. Make every environment identical in credentials

99. What should a developer generally do first when reproducing a reported bug?

A. Immediately rewrite the whole application
B. Establish reliable reproduction steps and expected versus actual behavior
C. Delete logs
D. Disable the failing feature permanently

100. Which design principle recommends that a module have one primary reason to change?

A. Single Responsibility Principle
B. Open/Closed Principle
C. Liskov Substitution Principle
D. Dependency Inversion Principle